A Twin Can Be Accurate and Still Wrong: Decision-Conditioned Fidelity for Network Digital Twins

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***Abstract*—Keeping a network digital twin (NDT) synchronised is costly: its telemetry competes with production traffic for the same resources. The literature budgets that cost in state error and twin age. We ask whether those are the right currency. A systematic mapping of 56 NDT papers finds one that links fidelity to the quality of the management decision the twin serves, while an adjacent goal-oriented communication literature has long established, for remote estimation, that decision-relevant information rather than raw freshness is what matters. We bridge the two on a closed-loop substrate where one fixed policy is driven by a twin degraded along four axes and compared against a perfect-information oracle over 97,200 seeded runs. Our central result is that state-space fidelity is structurally blind to model mismatch: where twin state error is exactly zero, up to 54% of decisions still differ from the oracle's. The resulting regret is small in absolute terms, 1–3% of the no-twin gap, but both incumbent metrics report a flawless twin. No fidelity metric predicts regret well: state error carries $R^2 = 0.42$ alone, rising only to 0.60 once decision type and load are known. Three of five pre-registered hypotheses failed; we report them.**

***Index Terms*—network digital twin, network and service management, synchronisation, fidelity, decision quality, telemetry cost, reproducibility**

I. Introduction

A network digital twin is valuable only insofar as something acts on it. In network and service management that something is a control loop: a scaling decision, an admission decision, a capacity split. Keeping the twin faithful to the physical network is not free, because the telemetry that carries state from the network to the twin consumes the very bandwidth, compute and energy that the managed services need. Operators therefore face a budgeting question every time they deploy a twin: how much fidelity must this twin have?

The literature answers a different question. Work on NDT synchronisation minimises the discrepancy between twin and physical state, or the age of the information the twin holds, and treats improvement in those quantities as the objective [1]–[3]. A dedicated metric vocabulary has grown around this framing, extending the Age of Information into an Age of Digital Twin [4]. What is largely missing is the connecting step: evidence that reducing state error or age actually improves the management decisions the twin exists to inform.

A systematic mapping of the field, reported in Section II, quantifies the gap. Of 56 NDT papers, 30 draw no link at all between fidelity and decision quality, 25 assert one qualitatively, and one establishes it quantitatively. Meanwhile a parallel literature on goal-oriented and semantics-aware communication has spent years establishing exactly this link for remote estimation [5]–[7], to the point of deriving a sampling threshold beyond which further sampling yields no decision improvement [8]. In our corpus these two literatures do not meet: no NDT paper cites any decision-conditioned freshness metric.

We do not claim to introduce decision-conditioned fidelity; that idea is established and we cite its lineage. We claim its absence from network digital twins has measurable consequences, and that network twins have three properties the remote-estimation results do not cover: a twin is a *model* rather than an estimate, extrapolating between updates with its own possibly wrong forward model; it covers its subject *partially*, under a shared telemetry budget; and it serves *several structurally different decisions at once* from one state representation.

We therefore quantify the disconnect between the two literatures across a 68-paper corpus; formalise decision-conditioned fidelity for network twins and measure it across 3,240 configurations and 97,200 seeded runs, isolating information quality by holding the policy fixed; and show that state-space metrics are structurally incapable of expressing model mismatch, exhibiting configurations where twin state error is exactly zero while most decisions diverge from the oracle's. Three of our five pre-registered hypotheses failed, and we report the failures alongside the confirmations.

II. Background and Related Work

We mapped publications from 2021 to 2026, retrieved by semantic search and screened against explicit inclusion criteria, yielding 56 NDT papers and 12 from the adjacent goal-oriented literature, each coded for the decision it serves, how it defines fidelity, its evaluation substrate, and whether it connects fidelity cost to decision quality (Fig. 1).

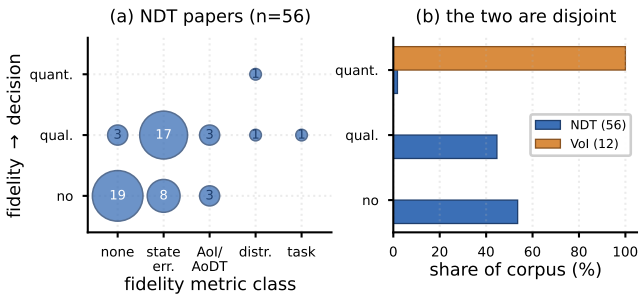


Fig. 1. Systematic mapping of 68 papers. (a) How the 56 network-digital-twin papers define fidelity, against whether they link it to decision quality; bubble area is the paper count. (b) The same linkage compared with the adjacent goal-oriented / value-of-information corpus. The two literatures are disjoint: no NDT paper in the corpus uses a decision-conditioned freshness metric.

Fidelity in the NDT corpus is overwhelmingly state-side: 25 papers measure state error or a similarity score, six use an age-based metric, and 22 never define it. Several optimise an age–accuracy tradeoff as the objective in its own right [1], [2], and one takes fidelity maximisation as the literal objective [3]. Surveys propose fidelity taxonomies as benchmarks [9] but leave them descriptive. The foundational framing of the NDT as an accurate real-time network model [10] and later work [11], [12] inherit the assumption: a more accurate twin is a better twin.

The closest work studies the effect of the twinning interval on network performance indicators [13]: it establishes that the knob matters, but measures throughput and latency rather than divergence from a reference decision, runs no perfect-information baseline, and derives no requirement. Elsewhere twins improve learned controllers, where the reported outcome is control performance, not a fidelity–decision relationship [14], [15].

The goal-oriented and value-of-information literature comes at the problem from the opposite side. The Age of Incorrect Information showed freshness metrics are blind to the role data plays [5]; the Goal-oriented Tensor was introduced because age-style metrics improve decision utility only indirectly [6]; scheduling by value of information beats distributing freshness evenly [7]; and recent work co-designs sampling with remote decision-making, deriving a threshold past which more data changes no decision [8]. All concern a transmitter and receiver tracking a source whose model the receiver knows — the assumption a network digital twin breaks.

III. Decision-Conditioned Fidelity

Let $s_t \in \mathbb{R}^K$ be the state of K managed service queues at control interval t, and let θ collect the network's service-rate parameters. A management policy π maps a state estimate and a parameter belief to an action, $a_t = \pi(\hat{s}_t, \hat{\theta})$. The twin supplies both arguments: it holds an estimate $\hat{s}_t$, refreshed from telemetry every Δ intervals with relative observation noise σ over a monitored fraction c of the queues, and between refreshes it advances that estimate using its own forward model parameterised by $\hat{\theta}$, which differs from θ by a relative error ε.

The *oracle* is the same policy applied to ground truth, $a_t^* = \pi(s_t, \theta)$; the *naive* reference acts on no state information. Holding π fixed across conditions isolates the effect of the information reaching the controller from policy adaptation, and makes the oracle a clean upper bound.

Three families of quantity follow. The incumbent, state-space metrics are the normalised state error $E_s = \langle \|\hat{s}_t - s_t\|_2 / \|s_t\|_2 \rangle_t$ and the age of digital twin A, the mean age of the freshest telemetry incorporated. The decision-level metrics are the action divergence $D_a = \langle \mathbb{1}[a_t \neq a_t^*] \rangle_t$, replaced by a normalised L_1 distance where the action space is continuous, and the regret $R = J(\pi \circ \text{twin}) - J(\pi \circ \text{oracle})$ in the operational objective J, which counts SLA-violating queue-intervals. We report the normalised regret $\tilde{R} = R/(J_{\text{naive}} - J_{\text{oracle}})$, so that $\tilde{R} = 0$ is oracle behaviour and $\tilde{R} = 1$ is the performance of having no twin. Values above one are meaningful and indicate a twin that is worse than none. Finally, telemetry cost M counts samples per interval. Every condition is evaluated under common random numbers, making R a paired difference.

The structural point that drives our main result is visible directly in the definitions. E_s is a function of the state pair $(\hat{s}_t, s_t)$ alone, whereas the policy reads both $\hat{s}_t$ and $\hat{\theta}$. Any twin error that lives in $\hat{\theta}$ rather than $\hat{s}$ is therefore invisible to E_s by construction, while remaining fully visible to the decision. Because a network digital twin is a model rather than an estimator, this is not a corner case but a routine failure mode.

IV. Methodology

The physical network is a discrete-time, vectorised simulation of $K = 12$ service queues sharing a capacity pool, with Markov-modulated Poisson arrivals over three states so load is non-stationary and autocorrelated; constant-rate arrivals would make extrapolation trivially easy. Utilisations are calibrated near 0.50, 0.75 and 0.92; each run covers 2,000 intervals after a 200-interval warm-up.

Three decisions of deliberately different structure are driven from the same twin: **DEC-A**, coarse discrete capacity scaling over five levels per queue; **DEC-B**, sharp-boundary admission control, binary per queue; **DEC-C**, a smooth continuous split on the simplex. Dropped arrivals cost 0.3 against 1.0 for an SLA violation, so admission control neither wins trivially by shedding load nor is barred from helping.

The twin is degraded along a full factorial of staleness $\Delta \in \{1, 2, 5, 10, 20, 50\}$, observation noise $\sigma \in \{0, 0.02, 0.05, 0.10, 0.25\}$, model mismatch $\varepsilon \in \{0, 0.05, 0.15, 0.30\}$ and coverage $c \in \{1.0, 0.75, 0.50\}$, crossed with three decisions and three load regimes: 3,240 cells, each run with 30 independent seeds derived by splitting a single master seed, for 97,200 runs totalling 7.5 minutes on two cores.

The design, hypotheses and analysis plan were registered before any full run executed; an amendment log records every

later change with its reason. Verification includes a degenerate-case test that a perfect twin ($\Delta = 1$, $\sigma = 0$, $\varepsilon = 0$, $c = 1$) yields exactly zero state error, action divergence and regret, passing to machine precision and establishing that the harness measures fidelity effects and nothing else. We also check queue conservation, monotonicity of E_s in each axis, and that the naive reference is strictly worse than the oracle in all nine decision–load combinations, without which normalised regret would have no denominator. Uncertainty is bootstrap 95% confidence intervals over seeds.

V. Results

A. A twin with zero state error can still be wrong

Fig. 2 isolates model mismatch by setting $\Delta = 1$, $\sigma = 0$ and $c = 1$, so the twin observes the true state perfectly at every interval and only its forward-model parameters are in error. The measured state error is exactly 0 in every one of these configurations, and the age of digital twin is likewise 0. Both incumbent fidelity metrics report a perfect twin.

The decisions tell a different story. Under DEC-A at $\varepsilon = 0.30$ the twin picks a different action from the oracle on 53.6% of intervals, with normalised regret 0.0109 (95% CI $[0.0081, 0.0137]$); under DEC-B the same mismatch gives 23.1% divergence and regret 0.0280 (CI $[0.0163, 0.0398]$), significant also at $\varepsilon = 0.15$. DEC-C shows exactly zero divergence at every mismatch level because its proportional-split policy allocates on estimated backlog alone and never reads the service-rate belief, so an error confined to $\hat{\theta}$ cannot reach its action. This is a property of the policy's parameter dependence rather than of the decision's smoothness: a DEC-C variant with the identical continuous action space, expressing need in service-time units and so consulting $\hat{\theta}$, diverges on up to 1.7% of intervals at zero state error — though its regret confidence interval still includes zero, so divergence here does not translate into cost. Three of the nine mismatch conditions survive Holm–Bonferroni correction across the family.

Regret magnitudes are modest here, because an otherwise perfectly synchronised twin recovers quickly. The finding is qualitative: a whole class of twin degradation is invisible to the metrics used to budget fidelity, and its impact depends on whether the policy consults the corrupted part of the model.

A reviewer may object that this is definitional: E_s compares states while the policy also reads model parameters, so a state metric must miss parameter error. That is precisely the argument, stated as such rather than dressed as a surprise; its force is empirical, not logical. The metric the objection calls inadequate is the one 25 of our 56 surveyed papers use to budget fidelity, six more using an age metric blind in the same way; a majority of intervals take a different action while both incumbent metrics report a flawless twin; and whether the blindness bites depends on whether the policy consults the corrupted part of the model, a fact about the controller that no twin-side metric can observe. The point sharpens for what-if analysis, the dominant use case in our corpus [13]: there the forward model is the product being consumed.

B. No fidelity metric predicts decision regret well

Across the full sweep, a bin-based fit of normalised regret on state error (quantile bins, no functional form assumed) explains $R^2 = 0.42$ of the variance. The age of digital twin does better at 0.60, and action divergence pooled across decisions worse at 0.18 — understated, since D_a carries different units for discrete and continuous action spaces; within decision type it reaches 0.27–0.39. Fitting within each decision and load stratum raises the state-error fit to 0.60, an increase attributable purely to context the metric does not carry.

The spread at matched fidelity makes the same point, with a caveat. Among the 265 configurations whose state error falls near 0.5, normalised regret ranges from 0.121 to 2.031, but part of that span is the normalising gap itself, which varies $19.5\times$ across the nine decision–load combinations. Within one combination the spread is smaller yet still material, reaching 0.65–0.97 for DEC-A at high load: a twin of identical measured fidelity costing between two-thirds and nearly all of the no-twin penalty.

That the age-based metric outperforms the accuracy-based one cuts against our own thesis and is worth stating plainly: staleness dominates here, and the freshness literature's currency tracks outcomes better than the accuracy literature's. Neither determines the decision consequence on its own.

C. Fidelity slack exists, but not as a general threshold

We pre-registered that regret would stay near zero over a broad region of state error before rising sharply. The evidence is axis-dependent. Along the observation-noise axis a genuine slack region exists: regret confidence intervals include zero up to a state error near 0.23 for all three decisions. Along the staleness and coverage axes no slack is observable at the granularity the system permits — the smallest increment, a twinning interval of two, already yields state error near 0.20 and regret between 0.07 and 0.12. There the relationship is graded, not threshold-like, and the strong form of our hypothesis is falsified.

We also pre-registered an ordering of slack width by decision type, expecting sharp-boundary decisions to demand the most fidelity. It is unsupported, and the question is ill-posed: the ordering differs by which degradation axis is varied, and on the one axis where slack is measurable the three values lie within 2% of each other. A single fidelity requirement per decision type is not well defined in our data, and on the model-mismatch axis it cannot be expressed at all, because state error is identically zero there.

D. A degraded twin can be worse than no twin

In 499 of 3,240 configurations (15.4%) normalised regret exceeds one, peaking at 1.77: the controller would have done better on a static allocation with no state information. A stale or mismodelled twin is a confident wrong signal, not a weak one, and the policy follows it.

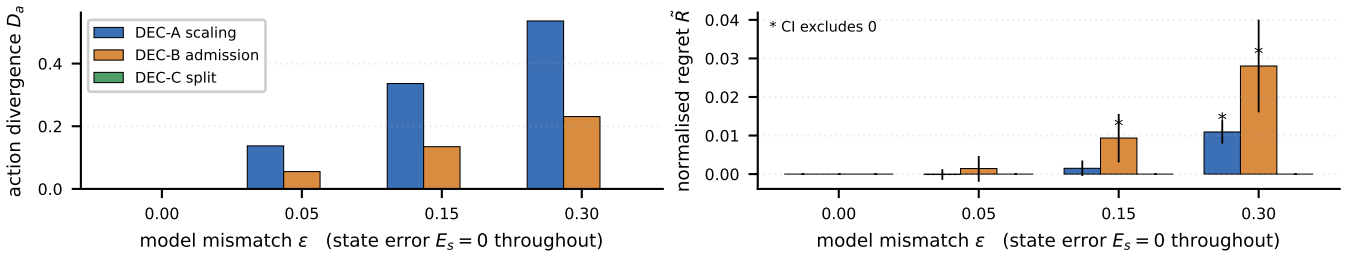


Fig. 2. State-space fidelity is blind to model mismatch. Across all bars the twin re-synchronises perfectly every interval, so its state error is *exactly zero*; only the forward-model parameters are wrong. Left: the fraction of intervals whose management action differs from the oracle's. Right: normalised regret with bootstrap 95% CIs; $*$ marks intervals excluding zero. DEC-C is immune only because its policy never consults the service-rate belief; a variant that does diverges (§V-A).

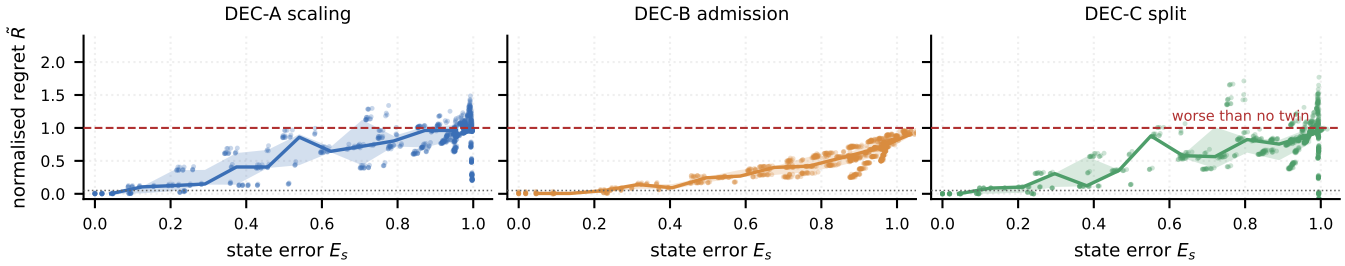


Fig. 3. Normalised regret against state error across all 3,240 configurations, by decision type. Points are configuration means; the line is the binned median with the interquartile band. The dashed line at $\tilde{R} = 1$ marks the performance of having no twin at all.

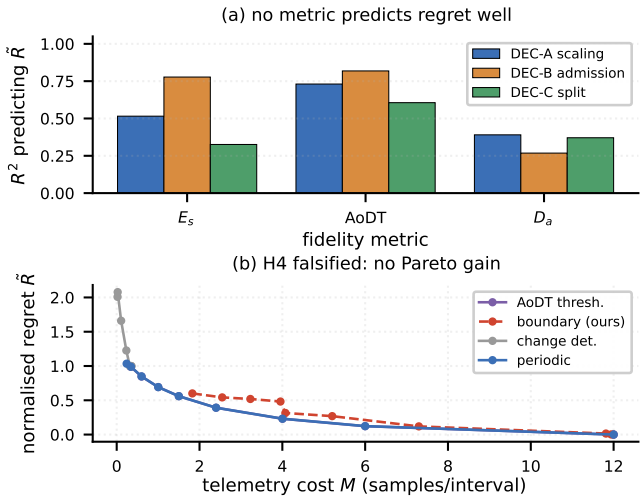


Fig. 4. (a) Variance in normalised regret explained by each fidelity metric, within decision type. (b) Telemetry cost against regret for four synchronisation policies (DEC-A, medium load). The proposed boundary-aware scheduler does not dominate periodic synchronisation.

E. Decision-aware scheduling did not help

Our fourth hypothesis proposed that a scheduler spending telemetry only when the twin's belief sits near a decision boundary would match periodic synchronisation at materially lower cost, with a pre-registered 30% saving as the success threshold. It was not met. Against periodic synchronisation the boundary-aware policy saves a median of 1.5% of telemetry at a regret target of 0.05 and 0.8% at 0.10, and at the looser target of 0.20 it is 22% *worse*. It also failed to reach the target at all in six of nine decision–load cells at $\tilde{R} \leq 0.05$ and five of nine at 0.10, where periodic synchronisation reached it in all nine; the medians above cover only the cells where it converged. Change-detection reached no target anywhere. H4 is falsified.

The failure is instructive. Boundary-aware scheduling must know whether the state sits near a decision boundary, and computes that from the twin's own belief — precisely what staleness corrupts. It is most confident no sample is needed exactly when its belief has drifted far enough to be wrong about where it stands; it appears to need an uncertainty estimate, not a point belief. Periodic and age-threshold scheduling also gave identical results: under full coverage all queues share one age, so the threshold degenerates into periodic sync. They are one baseline, not two.

VI. Threats to Validity

Construct. Normalised regret measures decision quality against one operational objective; a twin serving another could be budgeted differently. The oracle shares the policy's structure, so we measure the cost of degraded information, not of a suboptimal policy. **Internal.** Common random numbers remove environment variance but couple conditions; the degenerate-case test bounds harness error at zero. A pre-registered sensitivity analysis re-runs the headline conditions with Poisson arrivals, with $K = 24$, and with the dropped-arrival cost at 0.1 and 0.5: state error stays exactly zero, DEC-

A and DEC-B divergence rises to 0.60–0.79 and 0.25–0.53, DEC-C stays immune, and the drop cost changes nothing to four decimals. The mapping study coded most papers from abstracts rather than full texts, so category counts should be read as the field's dominant framing rather than a census. **External.** All results are simulation-based: no testbed, no operator traces, an abstracted topology. The numerical thresholds here should not be transferred to a deployed network; the structural claim about metric blindness follows from the definitions and does not depend on the simulator's realism. Mapping retrieval used one semantic search engine after direct scholarly-API access proved unavailable, so venue coverage is opportunistic. **Conclusion.** We report bootstrap intervals throughout and, where hypotheses failed, say so rather than reframing them; the mechanism proposed for the decision-structure differences is post-hoc and labelled as such.

VII. Conclusion

Network digital twins are budgeted in a currency — state error, age — that cannot express one of their characteristic failure modes. We showed configurations where twin state error is exactly zero while most management decisions diverge from the oracle's, because the error lives in the forward model rather than the state. No metric we measured predicts regret well, and in 15.4% of configurations a degraded twin is worse than no twin. Our proposed remedy failed, apparently needing an uncertainty estimate rather than a point belief. Fidelity requirements must be set against the decision, per decision, not inherited from a state-space budget.

Artifact. The simulator, pre-registration and amendment log, raw results, analysis and figure code accompany this paper; one command regenerates every number and figure in about ten minutes on two cores. A regression test guards per-cell seed determinism, after an adversarial self-review found Python's per-process hash randomisation had made the sweep non-reproducible. Results reproduce bit-identically across two environments, both yielding checksum `9f744f2268f4db0b`.

Appendix A
Disclosure Statement

Models used. All technical content — literature mapping, research question, simulator, experiments, analysis and this text — was generated by Anthropic's Claude (configured identifier `claude-opus-5`) as an autonomous agent in a Cowork session. The serving model may differ from the configured identifier; no other model was used.

Prompting strategy. One initial prompt, whose operative sentence read verbatim: *"The paper has to be produced with scientific rigor and look for novelty by proposing experiments, running them and analyzing the results, reproducibility is important."* It also asked for a plan, a literature search, a systematic mapping identifying gaps, a proposed topic with experiments, their execution and analysis, and a five-page paper on the supplied template. Four later interactions followed: a scoping round answered by multiple choice (direction deferred to a checkpoint; target the 1 September deadline; run autonomously with gates; pure Python substrate) and three gate decisions.

Framework and constraints. Agentic session with tool use: literature retrieval via a semantic scholarly connector with DOI verification against Crossref; Python 3.10/3.11 with NumPy, pandas, SciPy and Matplotlib; LaTeX with IEEEtran. Code was written and executed by the agent in two environments. Constraints: topic restricted to Digital Twins for Networks and Services; five-page IEEE format; simulation-only; two cores, no GPU; three approval gates; fixed deadline. A self-imposed rule required every citation to originate from a tool result and resolve against Crossref, enforced by a script that blocks the build.

Human roles and limits. K. Pereira and V. Vasquez composed and reviewed the prompts; J. Patinhas provided the hardware and issued them. The recorded human interaction was one initial prompt, one multiple-choice scoping round and three gate decisions, logged verbatim with timestamps in the artifact. No human edited any text, code, figure or reference, and all experiment execution was performed by the agent. There was no physical testbed, no real operator data and no human validation of the technical content.

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